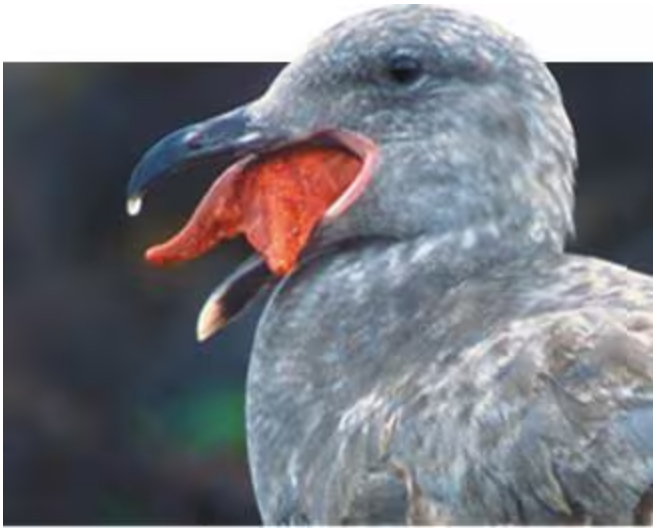
Chapter 41 Review

Campbell Biology, AP Edition, 12e

Chapter 41 Review



Summary of Key Concepts

Thinking in terms of surface-area-to-volume ratios, how do digestive processes influence the structure of different mammalian systems for digestion? **(BIG IDEA 2)**

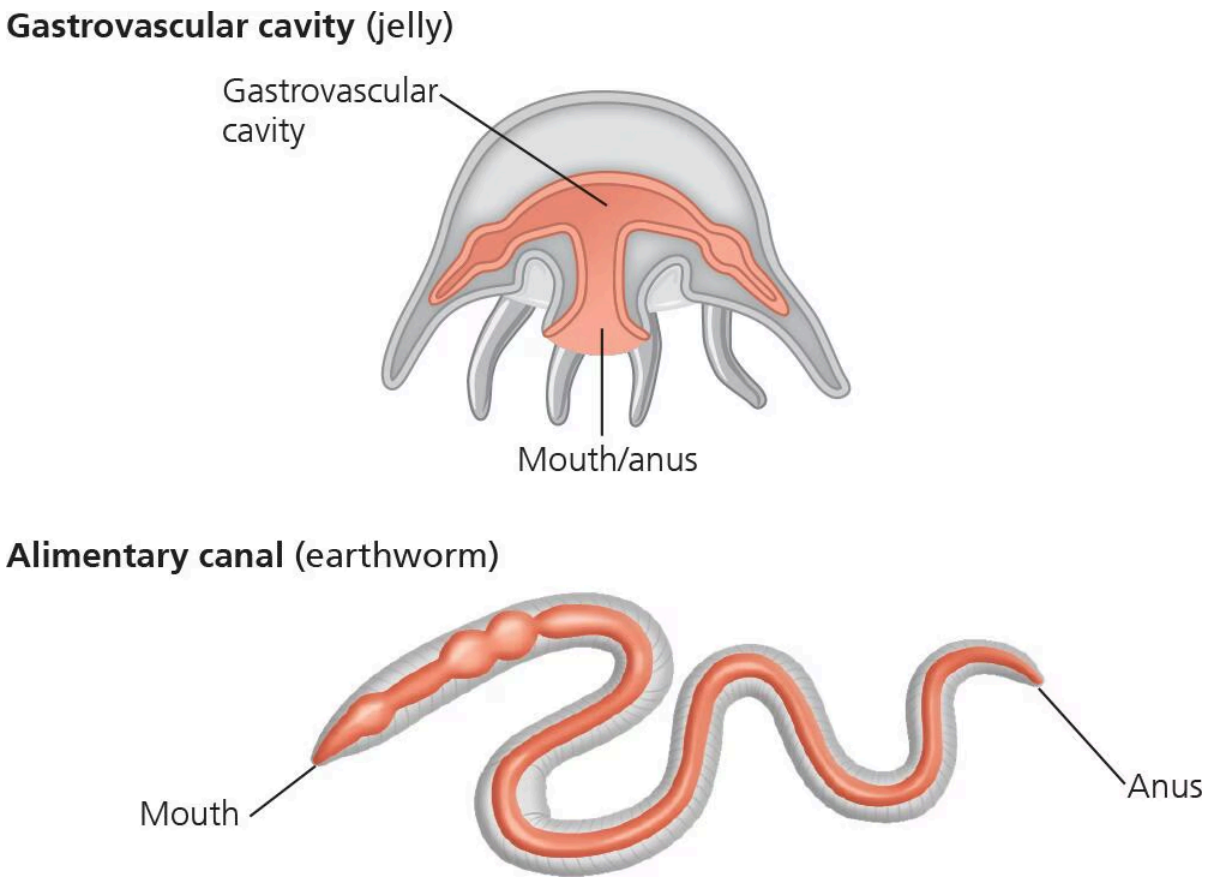
Concept 41.1: An animal's diet must supply chemical energy, organic building blocks, and essential nutrients

- Food provides animals with energy for ATP production, carbon skeletons for biosynthesis, and ****__essential nutrients__—nutrients that must be supplied in preassembled form. Essential nutrients include certain amino acids and fatty acids that animals cannot synthesize; ****__vitamins__, which are organic molecules; and ****__minerals__, which are inorganic substances.
- Animals have diverse diets. ****__Herbivores__ mainly eat plants; ****__carnivores__ mainly eat other animals; and ****__omnivores__ eat both. Animals must balance consumption, storage, and use of food.
- Malnutrition results from an inadequate intake of essential nutrients or a deficiency in sources of chemical energy. Studies of disease at the population level help researchers determine human dietary requirements.

How can an enzyme cofactor needed for an essential process be an essential nutrient for only some animals?

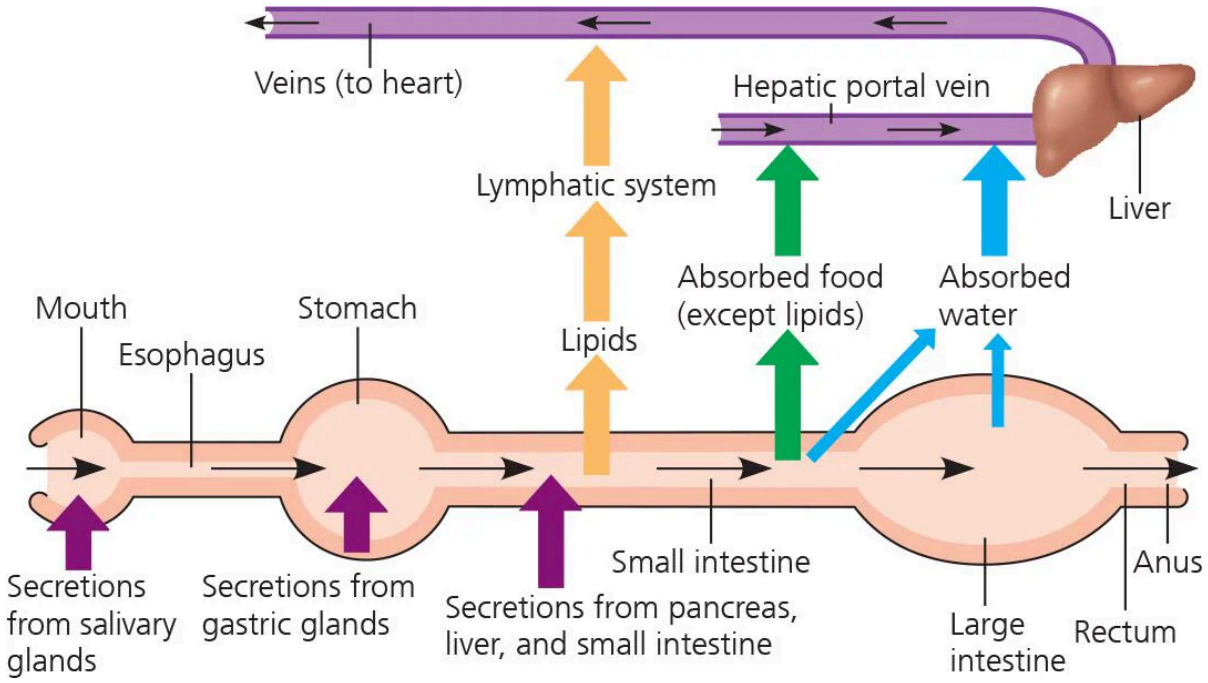
Concept 41.2: Food processing involves ingestion, digestion, absorption, and elimination

- Animals differ in the ways they obtain and ingest food. Many animals are ********__bulk feeders__, eating large pieces of food. Other strategies include **filter feeding**, ********__substrate feeding__, and ********__fluid feeding__.
- Compartmentalization is necessary to avoid self-digestion. In intracellular digestion, food particles are engulfed by phagocytosis and digested within food vacuoles that have fused with lysosomes. In extracellular digestion, which is used by most animals, enzymatic hydrolysis occurs outside cells in a ****__gastrovascular cavity__ or ****__alimentary canal__.



Propose an artificial diet that would eliminate the need for one of the first three steps in food processing.

Concept 41.3: Organs specialized for sequential stages of food processing form the mammalian digestive system



1. Food enters the mouth (with secretions from salivary glands).
2. Esophagus.
3. Stomach (with secretions from gastric glands).
4. Small intestine (with secretions from pancreas and liver); lipids travel to lymphatic system; absorbed foods except lipids travel to hepatic portal vein; to liver, and in veins to heart.
5. Large intestine (absorbed water travels to hepatic portal vein and follows path of other absorbed foods back to heart).
6. Rectum.
7. Anus.

What structural feature of the small intestine makes it better suited for absorption of nutrients than the stomach?

Concept 41.4: Evolutionary adaptations of vertebrate digestive systems correlate with diet

- Vertebrate digestive systems display many evolutionary adaptations associated with diet. For example, the assortment of teeth (dentition) generally correlates with diet. Also, many herbivores have fermentation chambers where mutualistic microorganisms digest cellulose. In addition, herbivores usually have longer alimentary canals than carnivores, reflecting the longer time needed to digest vegetation.

How does human anatomy indicate that our primate ancestors were not strict vegetarians?

Concept 41.5: Feedback circuits regulate digestion, energy storage, and appetite

- Nutrition is regulated at multiple levels. Food intake triggers nervous and hormonal responses that cause secretion of digestive juices and promote movement of ingested material through the canal. The hormones ******insulin** and ******glucagon** control the synthesis and breakdown of glycogen, thereby regulating glucose availability.
- Vertebrates store excess calories in glycogen (in liver and muscle cells) and in fat (in adipose cells). These energy stores can be tapped when an animal expends more calories than it consumes. If, however, an animal consumes more calories than it needs for normal metabolism, the resulting overnourishment can cause obesity.
- Several hormones, including leptin and insulin, regulate appetite by affecting the brain's satiety center.

Explain why your stomach might make growling noises when you skip a meal.